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Task Description:	Little is known about the mechanisms that underlie the greater biological effectiveness of high LET irradiation to promote solid cancer in experimental models. Our experimental data and that of others suggest that radiation carcinogenesis is a two-compartment problem: ionizing radiation can alter genomic sequence as a result of damage due to the interaction of energy and DNA, while altered phenotypes and multicellular interactions that may contribute to cancer remain poorly defined. As recently reviewed by Hanahan and Weinberg, the hallmarks of cancer occur within very specific microenvironments that promote malignancy (Hanahan and Weinberg, 2011). We hypothesize that both genetic and phenotypic changes contribute to radiation carcinogenesis. Radiation effects on critical phenotypes, if well understood, provide attractive targets for biological countermeasures because they represent processes that develop after radiation exposure and, unlike mutations, are amenable to intervention. We propose to study		

	three aspects of neoplasia: tissue stem cells, phenotype, and genomic instability as a function of HZE particle irradiation of non-malignant epithelial cells co-cultured with irradiated human fibroblasts and/or macrophages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is focused on developing a new experimental model of early stages of breast carcinogenesis representing critical cell interactions during radiation carcinogenesis. We contend that a more detailed mechanistic understanding of cell interactions that are affected by radiation will help to reduce the uncertainties in the age and gender dependence of cancer risks from the space radiation environment. We use a systems biology approaches to integrate molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the relevance of the cancer model to human data. The premise for these studies, that a pre-cancer niche formed by stromal cells promotes survival and malignant progression of initiated epithelial cells, is a new paradigm in carcinogenesis conceptualized by the PI. The concept was published in 2013 with colleagues in Nature Cancer Reviews. We propose that analysis of radiation effects on niche formation as a precursor to tumor microenvironment in this project will provide new data and a new perspective on preventing cancer. We hope to identify specific mechanisms that will serve as biomarkers that will allow comparisons among human tissues and will provide targets for countermeasures during space travel. Achieving these goals will also likely be useful to understanding and preventing risk from terrestrial radiation exposures.
	The proposed studies will examine mechanisms by which radiation affects specific human cell interactions and phenotypes that promote carcinogenesis. We will evaluate three epithelial phenotypes closely associated with early events in carcinogenesis: stem/progenitor self-renewal, genomic instability (GIN), and epithelial to mesenchymal transition (EMT). We will determine the effect of radiation dose and quality on these human epithelial cell phenotypes and then test whether irradiated human stromal cell interactions, i.e. macrophages and fibroblasts in co-cultures exacerbate radiation effects on premalignant phenotype. Introduction: The aim of this project is to develop an in vitro model to study the effect of densely ionizing radiation on normal mammary epithelial cells and their interactions with myeloid cells to identify mechanisms that contribute to increased carcinogenesis. Radiation is a well-established carcinogen, with the capacity to both initiate and promote the development of cancer. Several studies have demonstrated the role of radiation in initiating carcinogenesis by direct DNA damage that may lead to increased genomic instability and mutation. We have recently shown that radiation also has a considerable impact on the stromal compartment that contributes to the development of cancer (1-3). These in vivo experiments using irradiated mouse chimera models prove that host irradiation is sufficient to accelerate tumor development from an oncogenically primed epithelium and affects the malignant features of resulting tumors, even when epithelial cells have not been irradiated. These data and studies from others led us to propose the existence of a pre-cancer niche, formed by stromal cells, that promotes survival and malignant progression of initiated epithelial cells (4). One of the earliest events in the carcinogenesis process is the epithelial to mesenchymal transition (EMT). This trans-differentiation process occurs in early development stages and is characterized by a loss of epithelial polarity and cell-to-cell contacts accompanied by the acquisition of a mesenchymal and more motile phenotype. Our studies (2, 5-8) and others (9, 10) implicate EMT as a mechanism that results in dysplasia in early carcinogenesis while the EMT program is again employed in advanced tumors to promote invasion and metastasis (11, 12). Another important feature that epithelial cells must acquire during malignantly transformation and cancer development, is to escape from anti-proliferative signals. Transforming growth factor beta (TGFß) is thought to act as a tumor suppressor in large part because it suppresses epithelial proliferation (13). But TGFß is also a potent inducer of EMT, thus cells undergoing EMT-mediated dysplasia are also less proliferative, and epithelial cells must escape to the TGFß anti-proliferative effect for malignancy to progress (14). This paradox is a major focus of our proposed studies of the radiation quality dependence of intercellular interactions that promote neoplastic progression. Significance: Cancer is an important long-term risk for astronauts exposed to protons and high energy charged particles during travel and residence on other planets and the moon. There is no direct human data of the risk from the extended exposure to complex radiation fields. Our aim is to better understand the mechanisms by which high LET radiation increases carcinogenesis. We focus on the effects of radiation-induced TGFß as proof-of-principal of a microenvironment signal that may increase epithelial malignancy by multiple mechanisms. This will eventually lead us to the development of mechanism-based preventive therapies to decrease the risk of developing aggressive cancer after space travel. We also hope to identify specific mechanisms that could serve as biomarkers that will allow comparisons among human tissues and will provide targets for countermeasures during space travel. A near-term benefit of achieving these goals is that mechanistic understanding is also likely be useful to understanding health risks of terrestrial radiation exposures. Progress Aim 1: Characterize the radiation-induced signals that affect stem/progenitor self-renewal in human epithelial cells. In this aim, we proposed to evaluate radiation quality effects on the Let7c-reporter (Let7-R) and specific pathway inhibitors of stem/progenitor phenotypes and genomic instability (GIN). Since we eliminated the experiments using functional analysis of murine mammary stem/progenitor cell in vivo, we implemented a mammosphere assay that provides an in vitro surrogate of ‘stemmness’. MCF10A and Let7-R MCF10A cells treated with and without TGF beta and/or GSI were exposed to 600 MeV/amu Fe56 ions to evaluate the frequency of Let7c negative cells and their mammosphere forming ability. At 7 days post-radiation, the cells were FACS sorted based on the reporter into let7c negative and positive cells. Cells irradiated with high LET radiation replicated the published low LET response, i.e. Let7c-negative cells were increased in irradiated cultures exposed to TGF beta in a Notch dependent manner. Two mammosphere assays were conducted in parallel: the limiting dilution assay (LDA) and mammosphere forming efficiency (MFE) assay. The data from these studies were presented at Radiation Research Society meeting in Las Vegas in September. We compared the effect of 600 MeV/amu Fe56 versus X-rays on stem/progenitor phenotype and self-renewal in the progeny of irradiated MCF10A cells. Let7c negative cells increased in cultures irradiated with equal doses (50 cGy) of Fe56 ions (1.8 fold, $p < 0.05$) or X-rays (3.5 fold, $p<0.005$) compared to control cultures. This effect was Notch and

Task Progress:

TGFß-dependent. We next evaluated MCF10A functional self-renewal using a mammosphere assay. TGFß reduced mammosphere formation, consistent with its known inhibition of self-renewal and proliferation. The progeny of cells irradiated 5-6 days before with 10, 50, or 80 cGy of either high LET Fe56 ions or low LET X-rays showed decreased mammosphere forming capacity as measured by a quantitative limiting dilution assay. Notably, this decrement was blocked when cells were treated with TGFß. Thus, irradiation with either high or low LET similarly expand a rare population of Let7c negative MCF10A cells, which requires TGFß and Notch cooperation. These data suggest that although functional stem cell self-renewal is compromised in the daughters of irradiated cells, it is offset by TGFß in a Notch independent manner. As TGFß is a component of the tissue microenvironment, largely derived from stroma, these data support the concept of a stem cell niche, i.e. a specialized microenvironment that mediates self-renewal, that is altered by radiation exposure.

Aim 2: Determine whether macrophages promote malignant progression. The literature and a strong macrophage signature in our analysis of irradiated tissues microarray data indicate that macrophage phenotypes and distributions are affected by radiation (2). Here we will test the hypothesis that distinct subtypes of macrophages (M1 vs M2) act locally to release established monolayers of epithelial cells to undergo EMT and resistance to anti-proliferative signals. activated macrophages have the ability to secrete three major components that leads to the disruption of epithelial junctions and promote EMT: interferon gamma, which has been shown to induce lost of epithelial thigh junctions (15); metaloproteinases (i.e. MMP9) capable to degrade the extracellular matrix (16); and TGFß, which is required to induce radiation mediated EMT.

Monocytes isolated from peripheral blood mononuclear cells (PBMC) were obtained from healthy volunteer donors from NYC Blood Bank, and separated the monocytic fraction, from which we are able to differentiate M1 and M2 macrophages. The isolated monocytes are grown in media containing GM-CSF or M-CSF to respectively obtain M1 or M2 macrophage subtypes. M1 macrophages can be activated either with LPS and IFN gamma while M2 macrophages are then activated with TGFß. Thus we have five monocyte conditions: M0 (i.e. undifferentiated macrophages); M1, with and without activation; and M2, with and without activation. The distinct macrophage populations are analyzed using flow cytometry, staining for the following markers: CD45 (marker of hematopoietic cells), CD68 (marker of macrophages), CD80 (marker of M1 macrophages), and CD163 (marker of M2 macrophages). To assess EMT, we used semi-quantitative analysis of immunofluorescence for specific epithelial and mesenchymal markers in MCF10A non-malignant epithelial cells irradiated and treated with or without TGFß and/or cultured with the distinct macrophage subtypes or IFN gamma.

IFN gamma was used as proof of principle since it is a macrophage product that has been demonstrated to affect established epithelial junctions, which we predicted would elicit the EMT phenotype. As expected MCF10A that have established junctions did not undergo EMT in response to radiation and TGFß. The addition of IFN gamma (10 ng/ml) to irradiated cells treated with TGFß exhibited significantly decreased epithelial marker E-cadherin and increased translocation of ß-catenin from the cytoplasm to the nucleus. Preliminary experiments suggest that either M1 or M2 macrophages may cooperate with radiation-induced TGFß to induce EMT in normal mammary epithelial cells. We await the replicate of experiments in triplicate to assess the radiation quality dependence.

Our hypothesis is that EMT promotes acquisition of additional malignant characteristics. We determined that disrupted morphogenesis by EMT is maintained in MCF10A cells by increasing endogenous activation of TGFß, which could create a selective pressure to escape TGFß growth control and thus lead to increased proliferation evident in dysplasia. To test this idea, we speculated that TGFß resistant cells could be detected in these cultures as concomitant activation of TGFß signaling and proliferation. Given that IFN gamma elicits EMT, we tested whether it is is also both able to trigger resistance to TGFß's anti-proliferative effect. We observed a significant increase in a population of cells double positive for pSmad2 (readout of TGFß pathway activation) and Ki67 (a marker of proliferation) in those cultures treated with IFN gamma. Interestingly, this population was evident in the absence of irradiation.

Aim 3: Evaluate whether irradiated stromal cells increase the prevalence or persistence of preneoplastic phenotypes exhibited by irradiated epithelial cells. Our previous research demonstrated that EMT, GIN, and stem cell self-renewal occur in the daughters of irradiated cells as a function of radiation quality and fluence. In this aim, we proposed to simultaneously quantify EMT, GIN, and stem cell self-renewal using human breast epithelial cells co-cultured with normal or HZE irradiated fibroblasts.

The goal of experiments conducted at NSCRL-13A was to test how the EMT phenotype is affected by stromal cells. As reported last year, we encountered several technical problems with these studies initiated by Dr. Hatami. As with macrophages, fibroblast phenotypes are more complex as molecular markers are defined, thus we decided to delay further experiments while working out the functional co-cultures and assays, as described in Aim 2.

Accomplishments and future studies

The stem cell studies were presented as a poster at Radiation Research Society meeting by Dr. Hatami and I presented an overview of our NASA studies in a platform in the Presidential symposium. Dr. Gonzalez-Junca has efficiently generated distinct macrophage subtypes as well as characterized them using flow cytometry and quantitative real-time PCR analyzing for different macrophage activation markers (IL6, TNFa, MMP-9, and IL12 for M1 macrophages and ARG1, IL10, TGFß1, VEGF, and PDGF for M2 macrophages). We will complete the analysis of all the data generated in the previous NSRL14C run by the end of this year. The results from NSRL14A, B and C runs will be summarized in a poster presented by Dr. Gonzalez-Junca in the upcoming NASA Human Research Program Investigators' Workshop (HRP 2015). In the next runs during 2015 we will validate our hypothesis that macrophages cooperate with radiation-induced TGFß in mammary carcinogenesis by inducing EMT and generating cells resistant to TGFß anti-proliferative effect. In this sense we will also perform high-throughput gene-expression analysis using either microarrays or RNAseq to study the pathways that are regulated in the epithelial cells by co-culture with macrophages in different conditions. This will give us new information about candidate pathways and processes that are taking place in these early carcinogenesis events and will lead us to design rationale-based targeted therapies to prevent this process.

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